PHYS 580 Homework 3 (Chap.4)

All questions are worth 10 points, irrespective of complexity or length. Please submit all your solutions on Brightspace in PDF format, together with the source code (if any). Make sure to

a) discuss the physics and your results, as appropriate and/or directed,
b) include graphical output to illustrate the results,
c) include the source codes of your programs (at least the critical parts thereof),
d) describe briefly what your program does, and how it does it (algorithm), and
e) state the nature of the numerical approximation you used and demonstrate how you know that the approximation (with the parameters you used) is adequate for the problem you used it for.

Note the errata pages for the textbook: http://www.physics.purdue.edu/~hisao/book/www/errata.pdf
which are explained further on http://www.physics.purdue.edu/~hisao/book/www/errata.html

1. Problem E.2 (p.510)

   Calculate the period of a non-linear oscillator described by $\frac{d^2\theta}{dt^2} = -\sin\theta$ by numerically integrating $\int_0^{\theta_m} \frac{d\theta}{\sqrt{\cos\theta - \cos\theta_m}}$ for several values of the maximum angle θ_m, using the trapezoidal rule, Simpson's rule, or the Romberg integration method.

2. Problem 4.1 (p.100).

   Investigate the results obtained from the planetary motion program with different values of the time step. Show that for simulations of Earth, time steps greater than about $\Delta t = 0.01$ yr do not lead to satisfactory results. For such large time steps the orbits are not stable (i.e., not repeating). This is in accord with our general rule of thumb (Chapter 1) that the time step should be no larger than 1% or so of the characteristic time scale of the problem. In this case the characteristic time scale is the period of one orbit.

   Now use other initial speeds, both larger and smaller than the value of 2π (in AU units) which was needed for a circular orbit and again study the stability of the elliptical orbit to different values of the time step. Try values like 4 and 8 (in AU units). Do you see difference in the sensitivity to increasing time steps?

3. Problem 4.10 (p.113).

   Calculate the precession of the perihelion of Mercury, following the approach described in Section 4.3. *Only consider the correction coming from general relativity (ignore other Solar System planets).*

4. Problem 4.14 (p.118).

   Simulate the orbits of Earth and Moon in the solar system by writing a program that accurately tracks the motions of both as they move about the Sun. Be careful about (1) the different time scales present in this program, and (2) the correct initial velocities (i.e., set the initial velocity of Moon taking into account the motion of Earth around which it orbits).

   Also run your code for a hypothetical "Moon" that has either the same mass as Earth (e.g., binary planets), or a rather elliptical orbit.